Report - Version 12: Training Pipeline and Checkpoints

Introduction

Version 12 keeps the small decoder-only Transformer and public WikiText/BPE input pipeline introduced in Version 11. The external WikiText-103 Raw v1 source, training-only BPE tokenizer, target construction, two causal Transformer blocks, next-token objective, batch size, learning rate and 20-epoch budget are preserved.

The new concept is explicit training-state and checkpoint management. The notebook separates the latest state used to resume training from the best state selected by validation loss, then reconstructs a fresh final model from BEST before evaluating the test split.

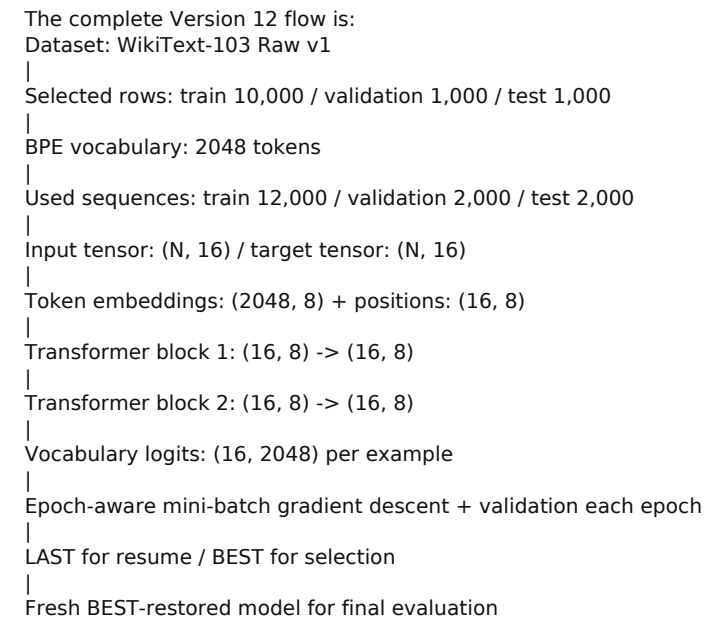
```
WikiText-103 text
|
BPE tokenizer trained on selected training text
|
16-token contexts + shifted targets
|
token + positional embeddings
|
Transformer block 1
|
Transformer block 2
|
vocabulary logits at every position
|
train to epoch 10 -> save LAST and BEST
|
restore LAST in a fresh model -> resume 11 to 20
|
restore BEST in a fresh final model
|
test evaluation + next-token prediction + greedy generation + attention inspection
```

Goal of Version 12

- preserve the complete Version 11 WikiText/BPE language-model experiment;
- track only the minimum state needed for resume: completed epoch, best epoch and best validation loss;
- save LAST every epoch so an interrupted run can continue from the correct next epoch;
- save BEST only when validation loss improves;
- simulate an interruption after epoch 10, restore LAST into a new model, and resume through epoch 20;
- restore another fresh model from BEST before test evaluation, prediction, generation and attention inspection;
- keep the implementation educational and explicit rather than adding production checkpoint infrastructure.

Architecture

The complete Version 12 flow is:



Continuity with Version 11

The notebook reads the same two WikiText training shards and official validation and test Parquet files used by Version 11. Empty rows are removed before deterministic subset sampling, and the original split boundaries remain unchanged. The BPE vocabulary is learned only from selected training text; validation and test never influence vocabulary construction.

Training-pipeline additions

- TransformerBlock and DecoderOnlyLanguageModel inherit from tf.Module only so their existing variables are checkpoint-trackable; the forward mathematics does not change.
- Training examples use shuffle_seed = SEED + epoch, allowing a restored run to continue with the shuffle associated with the correct next epoch.
- Validation loss is computed every epoch because BEST must be updated whenever validation improves.
- The test dataset is constructed earlier but is not evaluated until BEST selection is complete.

Trainable parameters

The model still contains 23 trainable TensorFlow variables and exactly 34,496 trainable parameters:

Component	Parameters
Token embeddings	2048 x 8 = 16384
Positional embeddings	16 x 8 = 128
Transformer block 1	800
Transformer block 2	800
Output weights	8 x 2048 = 16384
Total	34496

Each Transformer block still contributes 800 parameters:

Part	Shape	Params
Query projection	8 x 16	128
Key projection	8 x 16	128
Value projection	8 x 16	128
Attention output	16 x 8	128
LayerNorm 1 scale + shift	8 + 8	16
Feed-forward input	8 x 16	128
Feed-forward output	16 x 8	128
LayerNorm 2 scale + shift	8 + 8	16
Per block		800

Checkpoint state and semantics

Version 12 deliberately keeps checkpoint state minimal. completed_epoch records the most recently completed epoch; best_epoch records the epoch with the lowest validation loss; best_validation_loss stores that minimum. No optimizer object exists because Version 11 manual gradient updates are preserved.

State	Contents	Purpose
LAST	model + completed_epoch + best metadata	resume
BEST	model + best epoch/loss	selection
FINAL	fresh model restored from BEST	test + analysis

Loss, training and checkpoint results

Sparse softmax cross-entropy is still computed across all 16 target positions. Dataset losses are token-weighted across mini-batches. tf.GradientTape computes gradients for the same 23 trainable variables, sparse embedding gradients use scatter_sub, and dense gradients use assign_sub.

Setting	Value
Learning rate	1.0
Batch size	32
Seed	42
Interruption epoch	10
Total epochs	20
Training sequences	12,000 -> 375 batches
Validation sequences	2,000 -> 63 batches
Test sequences	2,000 -> 63 batches

Saved Version 12 execution

Measurement	Value
Initial train loss	7.626330
Initial validation loss	7.625952
Epoch 1 train / validation	6.030380 / 5.521137
Epoch 10 train / validation	5.068344 / 5.108189
Restored completed epoch	10
Resume begins at epoch	11
Epoch 20 train / validation	5.015265 / 5.082863
BEST epoch	18
BEST validation loss	5.074273
Final BEST-restored train loss	4.986566
Final BEST-restored validation loss	5.074273
Test loss	5.105811
Uniform-vocabulary baseline	7.624619

The small difference between the epoch-20 validation loss 5.082863 and the lower BEST loss at epoch 18 5.074273 makes the new lesson concrete: LAST and BEST are not guaranteed to identify the same model. LAST answers where training ended and can resume; BEST answers which observed model validation selected.

Inspection of next-token probabilities and attention

For the unseen test context "Little Gidding is the fourth and final poem", the largest final-position probabilities are comma 0.1380 period 0.0794 and 0.0469 in 0.0345 apostrophe 0.0256 was 0.0258 and of 0.0289 The true next token is of with probability 0.0289 Block 1 final-position attention is almost uniform near 0.0625. Block 2 is more selective, with large saved weights including G = 0.3228 f = 0.1224 the = 0.0520 and L = 0.2162 Both final-position rows sum to 1.000000, and the full tests confirm zero attention to future positions.

Autoregressive generation

Generation remains identical in method to Version 11: the prompt is encoded with the same BPE tokenizer, the most recent 16 token identifiers form the context, and greedy decoding repeatedly chooses the highest-probability next token. The only Version 12 change is that generation uses final_model restored from BEST.

Saved prompt:

Little Gidding is the fourth and final poem

Saved decoded generation:

Little Gidding is the fourth and final poem , and the Sess , and the Sess , and the Sess , and the Sess , and the Sess , and the Se

The run adds exactly 40 BPE tokens. The repeated pattern is consistent with the deliberately tiny model, 16-token context and greedy decoding. Whitespace pre-tokenization can also produce spaces before punctuation; this is a tokenizer-formatting effect rather than a Transformer error.

Included checks

- the tokenizer vocabulary size is exactly 2048;
- training, validation and test input/target tensors have the configured 16-token sequence length and capped split sizes;
- the first target sequence matches the first input shifted by one token;
- model output shape is (1, 16, 8), logits are (1, 16, 2048), two independent Transformer blocks remain, and the parameter count is exactly 34,496;
- LAST and BEST checkpoint files exist;
- restored_epoch is exactly 10 and resumed completed_epoch reaches 20;
- selected_best_epoch is exactly the validation-selected best epoch 18;
- final_model is a distinct object restored from BEST;
- final validation loss matches the validation loss stored in BEST within numerical tolerance;
- all reported losses are finite, final train/validation improve over their initial values, and test loss remains below the random baseline;
- causal attention rows sum to 1 and future entries are zero;
- the inspected next-token probability distribution is finite and sums to 1;
- autoregressive generation adds exactly 40 new BPE tokens;
- the notebook finishes with All checks passed.

What Version 12 teaches

Version 12 is the first stage in the project where model training becomes explicitly resumable and validation-based model selection is separated from the latest in-memory parameters. The conceptual progression is:
Version 11 WikiText/BPE model
-> reusable training epoch
-> deterministic epoch-specific shuffling
-> LAST checkpoint for continuity
-> interruption at epoch 10
-> fresh model restored from LAST
-> resume at epoch 11 through epoch 20
-> BEST checkpoint selected by validation loss
-> fresh final model restored from BEST
-> one final test evaluation and post-training analysis

The important distinction from Version 11 is not a new model equation. Dataset representation, BPE tokenization and Transformer mathematics stay intentionally unchanged so the effect of explicit training state, resume and model selection can be isolated directly.

Limitations

- only controlled subsets of WikiText-103 are used;
- the model remains intentionally tiny with 8-dimensional representations, two Transformer blocks and one attention mechanism per block;
- context length remains fixed at 16 BPE tokens;
- the whitespace-based BPE configuration is intentionally simple and can place spaces before punctuation;
- greedy decoding remains deterministic and can amplify repetitive patterns;
- the runtime-v12 directory is deliberately deleted and recreated at the start of each Run All, so persistence across separate Kaggle sessions is outside this educational version;
- checkpoint state is intentionally minimal and does not include optimizer state because the notebook still uses manual gradient updates;
- there is still no learning-rate schedule, dropout, mixed precision or larger-scale training infrastructure.

Conclusion

Version 12 successfully adds explicit checkpoint-based training continuity and validation-based model selection to the Version 11 public-data subword language model without changing its core architecture. The saved execution demonstrates the full intended behavior: training stops at epoch 10, LAST restores a fresh model and resumes at epoch 11, training reaches epoch 20, BEST identifies epoch 18 as the lowest-validation model, and a separate final model restored from BEST produces the final training, validation and test measurements before prediction, generation and attention inspection. This gives the project a clean training-state foundation for later optimization and scaling features.